30-Minute Student Task

Build Your Own Rule-Based Python Chatbot

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Programming Language	Python

Assessment

A simple rule-based conversational chatbot was developed in Python. The chatbot accepts user input continuously, recognizes greetings and different questions, responds accordingly, and exits when the user types bye or exit.

Demonstration Evidence

The following pages contain the Python implementation and the terminal execution.

Python Chatbot Implementation

Program code implementation in python:

```
name = input("Enter your name: ")

print("Hello", name, "!")
print("I am a simple chatbot. Type 'bye' or 'exit' to stop.")

while True:
    message = input("You: ").lower().strip()

    if "hello" in message or "hi" in message:
        print("Bot: Hello", name, "! How are you?")
    elif "how are you" in message:
        print("Bot: I am doing great! Thanks for asking.")
    elif "what is python" in message:
        print("Bot: Python is a popular programming language.")
    elif "what can you do" in message:
        print("Bot: I can answer simple questions and chat with you.")
    elif "bye" in message or "exit" in message:
        print("Bot: Goodbye", name, "!")
        break
    else:
        print("Bot: Sorry, I don't understand that. Try asking something
else.")
```

Chatbot Execution and Test Output

Terminal output showing greeting, question handling, an unknown message, and the exit command.

```
PS C:\Users\annalakshmi\Desktop\python_task> python chatbot.py
Enter your name: Annalakshmi
Hello Annalakshmi !
I am a simple chatbot. Type 'bye' or 'exit' to stop.
You: hello
Bot: Hello Annalakshmi ! How are you?
You: how are you
Bot: I am doing great! Thanks for asking.
You: what is python
Bot: Python is a popular programming language.
You: difference between python and java
Bot: Sorry, I don't understand that. Try asking something else.
You: bye
Bot: Goodbye Annalakshmi !
PS C:\Users\annalakshmi\Desktop\python_task>
```